

Chapter 16: Molluscs

A Significant Space

- The evolution of a **coelom** (fluid-filled space within the mesoderm) was a major step in the evolution of larger and more complex forms.
- Unlike pseudocoelomates where organs lie loose, in coelomates, organs are suspended by **mesenteries** (mesodermal membranes).
- Mesenteries provide ideal locations for blood vessel networks and allow the alimentary canal to become more muscular and specialized without interfering with other organs.
- In many coelomates, the coelom serves as a **hydrostatic skeleton**, but in molluscs, the heavy shell restricts this function, and the coelom is relatively small.

Molluscs

- **Mollusca** is one of the largest animal phyla, with over 90,000 living species.
- They are **coelomate lophotrochozoan protostomes** with spiral mosaic cleavage and schizocoelous coelom formation.
- The ancestral larval stage is a **trochophore**.
- The group is extremely diverse, ranging from microscopic forms to giant squids (**Architeuthis**).

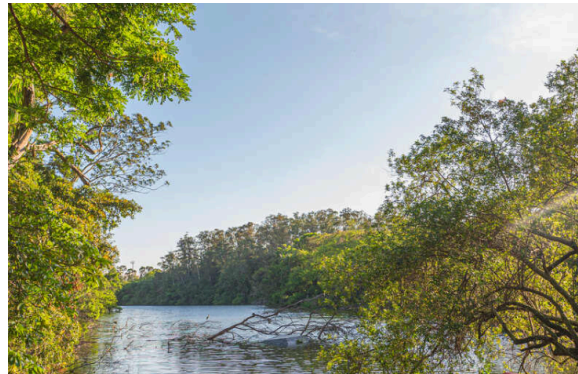


Figure 1: Molluscs: a diversity of life-forms. The basic body plan of this ancient group has become variously adapted for different habitats. A, A chiton (*Tonicella lineata*), class Polyplacophora. B, A marine snail (*Calliostoma annulata*), class Gastropoda. C, A nudibranch (*Chromodoris* sp.) class Gastropoda. D, Pacific giant clam (*Panope abrupta*), with siphons to the left, class Bivalvia. E, An octopus (*Octopus briareus*), class Cephalopoda, forages at night on a Caribbean coral reef.

- Molluscs are found in nearly all habitats: marine (tropics to polar, surface to abyss), freshwater, and terrestrial (snails and slugs).
- Humans exploit molluscs for food, pearls, and buttons, but some are pests (e.g., shipworms, garden snails, intermediate hosts for parasites).
- The phylum includes several classes with varying degrees of diversity and distinct synapomorphies, as shown in the cladogram.

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Figure 2: Cladogram showing hypothetical relationships among classes of Mollusca. Synapomorphies that identify the various clades are shown, although a number of these have been modified or lost in some descendants.

Form and Function

- The generalized mollusc body plan consists of two main portions: the **head-foot** (active area) and the **visceral mass** (metabolic area).

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Figure 3: Generalized mollusc. Although this construct is often presented as a “hypothetical ancestral mollusc (HAM),” most experts now reject this interpretation. Such a diagram is useful, however, to facilitate description of the general body plan of molluscs.

- Two folds of skin form the **mantle**, which encloses the **mantle cavity** containing gills (**ctenidia**) or lungs and secretes the **shell**.

Head-Foot

- Most molluscs have a well-developed head with mouth and sensory organs.
- **Radula**: A rasping, protrusible, tonguelike organ found in all molluscs except bivalves and most solenogasters.
 - It consists of a ribbonlike membrane with rows of tiny teeth pointing backward.
 - It moves over a supporting cartilage called the **odontophore**.
 - Used for feeding (scraping, piercing, tearing, cutting) and conveying food to the digestive tract.



Figure 4: A, Diagrammatic longitudinal section of a gastropod head showing a radula and radula sac. The radula moves back and forth over the odontophore cartilage. B, Radula of a snail prepared for microscopic examination.

- **Foot:** The chief locomotor organ, usually ventral and solelike.
 - Adapted for creeping, attachment (limpets), burrowing (bivalves), or jet propulsion (cephalopods).
 - Can be extended hydraulically by engorgement with blood.

Visceral Mass

- **Mantle and Mantle Cavity:**
 - The mantle protects soft parts and creates the mantle cavity.
 - The mantle cavity houses respiratory organs (gills/lungs) and receives products from digestive, excretory, and reproductive systems.
 - In aquatic forms, water currents bring oxygen/food and flush out wastes.
 - **Ctenidium** (gill): Consists of a central axis with leaf-like filaments. Blood flow is opposite to water flow (**countercurrent exchange**).

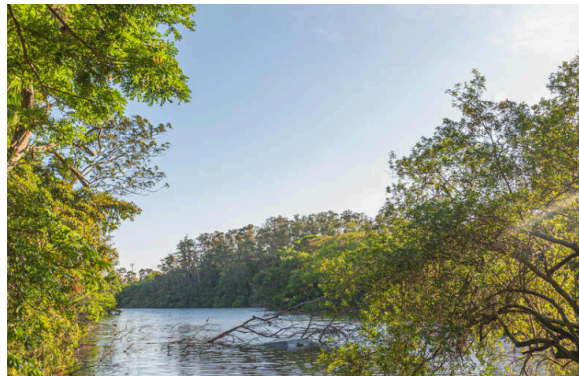


Figure 5: Primitive condition of mollusc ctenidium. Circulation of water between gill filaments is by cilia, and blood diffuses through the filament from the afferent vessel to the efferent vessel. Black arrows are ciliary cleansing currents.

- **Shell:** Secreted by the mantle and typically has three layers:
 - **Periostracum:** Outer organic layer (conchiolin), protects against erosion.
 - **Prismatic layer:** Middle layer of densely packed calcium carbonate prisms.
 - **Nacreous layer:** Inner layer, secreted continuously, forming the iridescent “mother-of-pearl”.
 - Pearls are formed when a foreign object becomes lodged between the shell and mantle and is covered with nacre.



Figure 6: A, Diagrammatic vertical section of shell and mantle of a bivalve. The outer mantle epithelium secretes the shell; the inner epithelium is usually ciliated. B, Formation of pearl between mantle and shell as a parasite or bit of sand under the mantle becomes covered with nacre.

Internal Structure and Function

- **Circulatory System:** Open circulatory system (except cephalopods) with a pumping heart, vessels, and blood sinuses. Less efficient than closed systems.
- **Excretion:** Most have a pair of kidneys (**metanephridia**) opening into the coelom via a **nephrostome**.
- **Nervous System:** Paired ganglia (cerebral, pleural, pedal, visceral) with nerve cords.
- **Reproduction:**
 - Most are dioecious; some hermaphroditic.
 - **Trochophore** larva: Free-swimming, top-shaped larva with a circlet of cilia (**prototroch**); ancestral for molluscs and annelids.

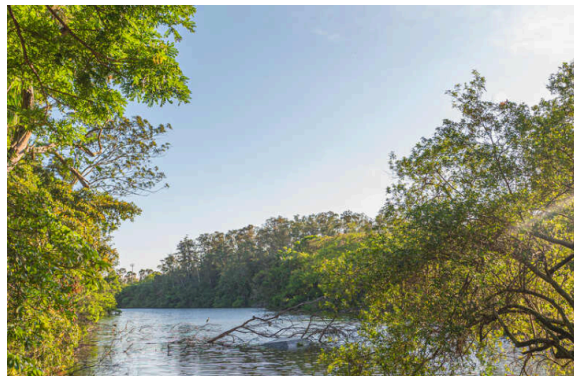


Figure 7: A, Generalized trochophore larva. Molluscs and annelids with an ancestral pattern of embryonic development have trochophore larvae, as do several other phyla. B, Trochophore of a Christmas tree worm, *Spirobranchus spinosus* (Annelida).

- **Veliger** larva: A uniquely molluscan larval stage with the beginnings of a foot, shell, and mantle, and ciliated velar lobes for swimming.



Figure 8: Veliger of a snail, *Pedicularia*, swimming. The adults are parasitic on corals. The ciliated process (velum) develops from the prototroch of the trochophore.

Classes of Molluscs

Class Caudofoveata

- Wormlike, marine, burrowing animals.
- Lack a shell but have calcareous scales/spicules.
- Have an oral shield and a radula.

Class Solenogastres

- Wormlike, marine, bottom-dwellers (often feed on cnidarians).
- Usually lack radula and gills.
- Foot is represented by a midventral pedal groove.

Class Monoplacophora

- Thought extinct until discovery of *Neopilina* in 1952.
- Small, with a single low, rounded shell and creeping foot.
- Exhibit **pseudometamerism**: serial repetition of organs (gills, muscles, nephridia).
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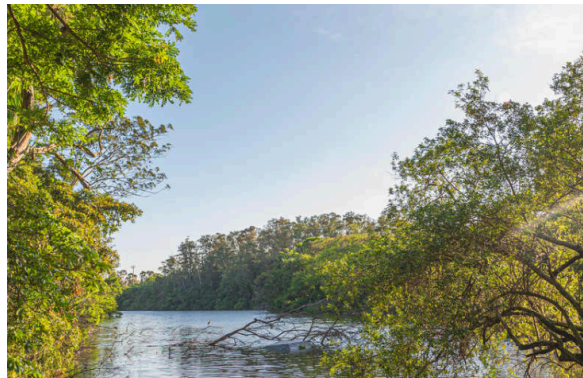


Figure 9: *Neopilina*, class Monoplacophora. Living specimens range from 3 mm to about 3 cm in length. A, Ventral view. B, Dorsal view.

Class Polyplacophora: Chitons

- Dorsoventrally flattened with a convex dorsal surface bearing **seven or eight articulating limy plates**.
- Head and sensory organs reduced; **esthetes** (photosensitive structures) pierce the plates.

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Figure 10: Anatomy of a chiton (class Polyplacophora). A, Longitudinal section. B, Transverse section. C, External ventral view.

- Prefer rocky intertidal surfaces; cling tightly with a broad foot.
- Mantle forms a girdle around the plates.

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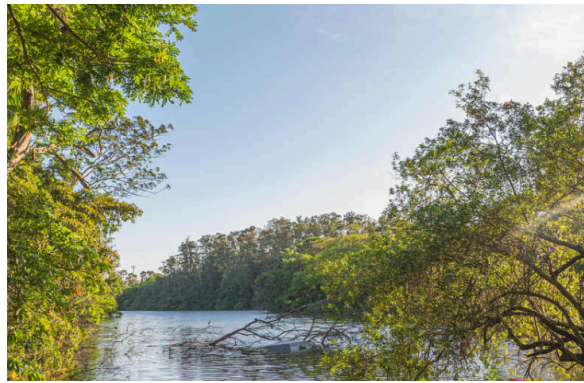


Figure 11: Mossy chiton, *Mopalia muscosa*. The upper surface of the mantle, or “girdle,” is covered with hairs and bristles, an adaptation for defense.

- Gills are numerous and suspended in the mantle grooves along the sides of the foot.

Class Scaphopoda

- **Tusk shells** or tooth shells.
- Benthic marine; slender body covered by a mantle and a tubular shell open at both ends.
- Foot protrudes through the larger end to burrow.
- Gills are absent; gas exchange occurs in the mantle.
- Food is caught by **captacula** (ciliated, adhesive tentacles).

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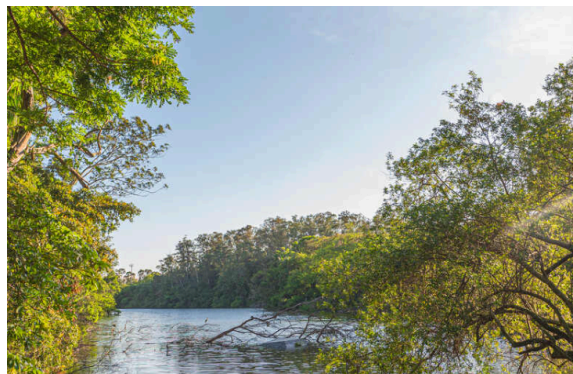


Figure 12: The tusk shell, *Dentalium* (class Scaphopoda). A, It burrows into soft mud or sand and feeds by means of its prehensile tentacles (captacula). B, Internal anatomy of *Dentalium*.

Class Gastropoda

- Largest and most diverse class (snails, limpets, slugs, nudibranchs, etc.).
- Shell, when present, is **univalve** (one piece), coiled or uncoiled.
- Shell features: **apex**, **whorls**, **columella** (central axis), **aperture**, **operculum** (cover).
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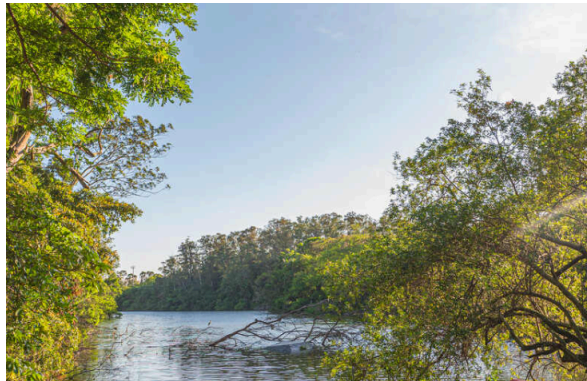


Figure 13: Shell of the whelk *Busycon*. A and B, *Busycon carica*, a dextral, or right-handed, shell. C, *B. contrarium*, a sinistral, or left-handed, shell.

Torsion

- **Torsion** is a developmental process where the visceral mass rotates 180 degrees counterclockwise relative to the head.
- Moves the anus and mantle cavity to the anterior position, above the head.
- **Ontogenetic torsion**: The developmental sequence.
- **Evolutionary torsion**: The evolutionary history.
- Advantages: Head can be withdrawn into shell; sensory organs in mantle cavity face forward.
- Disadvantage: **Fouling** (wastes washing back over gills and head).
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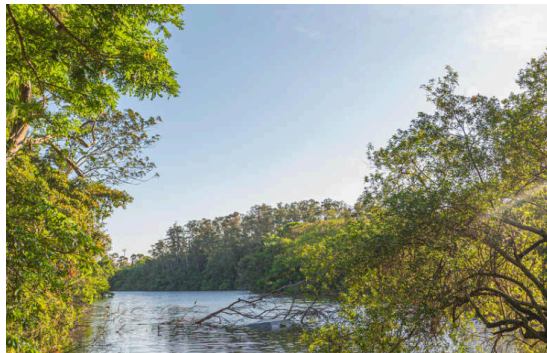


Figure 14: Ontogenetic torsion in a gastropod veliger larva. The sequence shows the rotation of the shell and viscera relative to the head and foot.

Coiling

- **Coiling** is the spiral winding of the shell and visceral mass (separate from torsion).
- Evolution from **planospiral** (all whorls in one plane) to **conispiral** (whorls to the side).
- Conispiral shape caused unbalance, solved by shifting shell upward and posteriorly.
- This shift pressed on the right side of the mantle cavity, leading to the loss of the right gill, auricle, and kidney (**bilateral asymmetry**).

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Figure 15: Evolution of shell in gastropods. A, Earliest coiled shells were planospiral. B, Conispiral shape for better compactness. C and D, Shell shifted upward and posteriorly for better weight distribution.

Feeding Habits

- Varied feeding: herbivorous grazers (abalone), scavengers, carnivores, borers, ciliary feeders.

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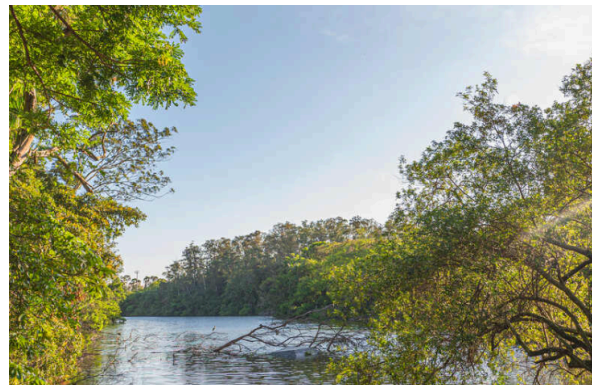


Figure 16: A, Red abalone, *Haliotis rufescens*. This huge, limpetlike snail is prized as food. B, Moon snail, *Polinices lewisii*. A common inhabitant of West Coast sand flats, the moon snail is a predator of clams and mussels.

- **Conus** snails use a modified radula tooth as a harpoon with toxic venom (**conotoxins**) to capture prey.

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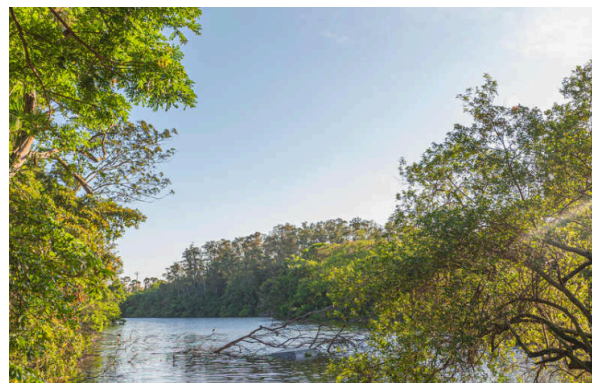


Figure 17: *Conus* extends its long, wormlike proboscis (A). When a fish attempts to consume this tasty morsel, the *Conus* stings it in the mouth and kills it. The snail engulfs the fish with its distensible stomach (B).

Internal Form and Function

- Respiration: Typically by a **ctenidium** in the mantle cavity.
- To avoid fouling, water often enters the left side and exits the right side (carrying wastes).
- Evolution of gills: Loss of right gill and half of the left gill filaments to improve efficiency.



Figure 18: Evolution of ctenidia in gastropods. A, Primitive condition with two ctenidia. B, Condition after one ctenidium had been lost. C, Derived condition found in most marine gastropods, in which filaments on one side of remaining gill are lost, and axis is attached to the mantle wall.

- **Pulmonates** lack gills; the vascularized mantle wall functions as a **lung**, opening via a **pneumostome**.

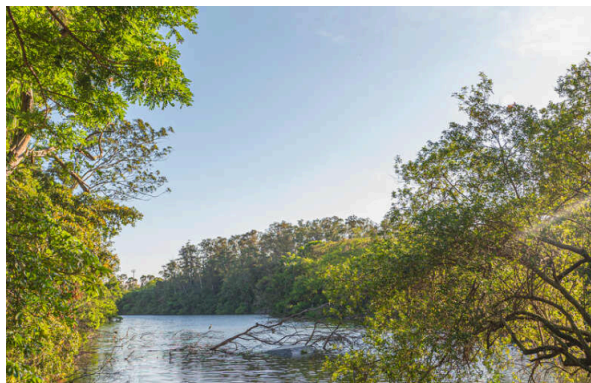


Figure 19: Anatomy of a pulmonate snail.

- Reproduction: Monoecious or dioecious; internal fertilization in most; eggs often laid in capsules.

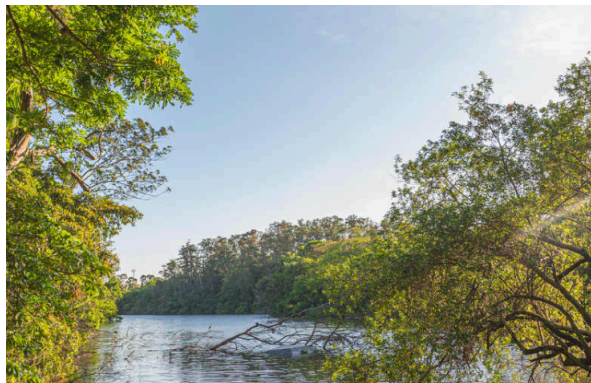


Figure 20: Eggs of marine gastropods. A, The wrinkled whelk, *Nucella lamellosa*, lays egg cases resembling grains of wheat. B, Egg ribbon of a dorid nudibranch.

Major Groups

- **Prosobranchs:** Mostly marine; mantle cavity anterior; gills in front of heart; operculum usually present. Examples: abalones, whelks, periwinkles.
 - Some have holes in the shell for excurrent water (e.g., keyhole limpets) to avoid fouling.

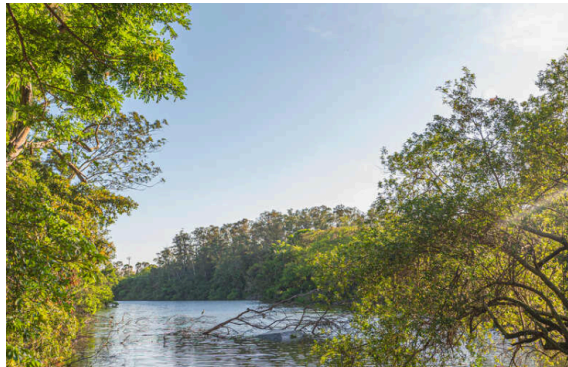


Figure 21: A, *Diodora aspera*, a gastropod with a hole in its apex through which water leaves the mantle cavity. B, Flamingo tongues, *Cyphoma gibbosum*, are showy inhabitants of Caribbean coral reefs.

- **Opisthobranchs:** Marine (sea slugs, sea hares, nudibranchs); show partial or complete **detorsion** (anus to right or posterior); shell reduced or absent.
 - **Sea hares (*Aplysia*)** have large parapodia and release protective ink.



Figure 22: A, The sea hare, *Aplysia dactylomela*, crawls and swims across a tropical seagrass bed, assisted by large, winglike parapodia. B, When attacked, sea hares squirt a copious protective secretion from their “purple gland” in the mantle cavity.

- **Nudibranchs** are carnivorous, often brightly colored, and may use hijacked nematocysts from cnidarian prey for defense.



Figure 23: *Phyllidia ocellata*, a nudibranch. Like other *Phyllidia* spp., it has a firm body with dense calcareous spicules and bears its gills along the sides, between its mantle and foot.

- **Pulmonates:** Land and freshwater snails/slugs; mantle cavity modified into a lung; monoecious.
 - Land forms have two pairs of tentacles (posterior pair bears eyes).

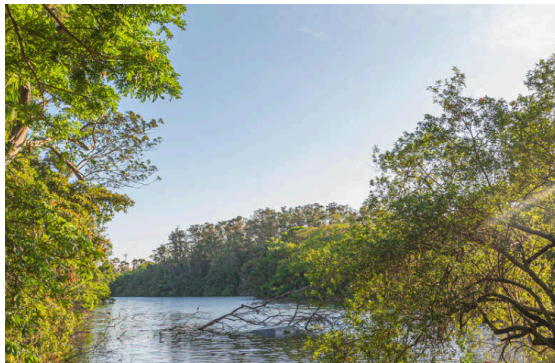


Figure 24: A, Pulmonate land snail. Note two pairs of tentacles; the second, larger pair bears the eyes. B, Banana slug, *Ariolimax columbianus*. Note pneumostome.

Class Bivalvia (Pelecypoda)

- Mussels, clams, scallops, oysters, shipworms.
- Two shells (**valves**) held by a hinge ligament; no head or radula; mostly filter feeders.
- Invasive species like the **zebra mussel** (*Dreissena polymorpha*) cause significant ecological and economic damage.

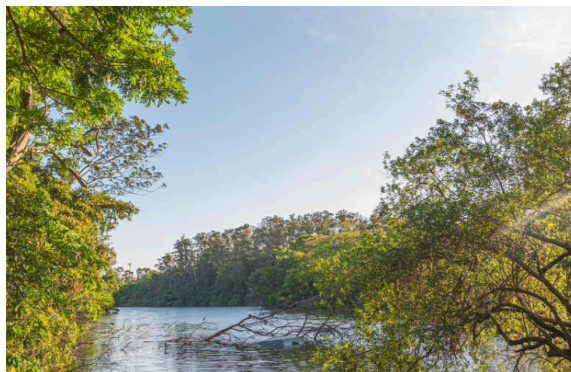


Figure 25: Bivalve molluscs. A, Mussels, *Mytilus edulis*, occur in northern oceans around the world. B, Scallops (*Chlamys opercularis*) swim to escape attack by starfish (*Asterias rubens*).

Form and Function

- **Shell:** Valves drawn together by **adductor muscles** working against the hinge ligament. **Umbo** is the oldest part.

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Figure 26: *Tagelus plebius*, stubby razor clam (class Bivalvia). A, External view of left valve. B, Inside of right shell showing scars where muscles were attached. C and D, Sections showing function of adductor muscles and hinge ligament.

- **Body:** Visceral mass suspended dorsally; muscular foot attached anteroventrally; ctenidia hang on each side.
- **Gills:** Modified for filter feeding. Filaments lengthened and folded to form **lamellae** with water tubes inside.

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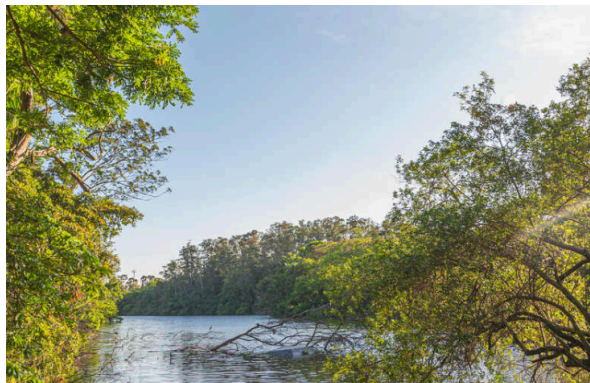


Figure 27: Sections through a bivalve shell and body, showing relative positions of visceral mass and foot. Evolution of bivalve ctenidia: By a great lengthening of individual filaments, ctenidia became adapted for filter-feeding.

- **Siphons:** Posterior mantle edges modified into incurrent and excurrent siphons for water flow, allowing burrowing.



Figure 28: Adaptations of siphons in bivalves. A, In northwest ugly clams *Entodesma saxicola*, incurrent and excurrent siphons are clearly visible. B to D, In many marine forms the mantle is drawn out into long siphons.

- **Locomotion:** Foot extended, engorged with blood to anchor, then contracted to pull body forward. Scallops swim by clapping valves.

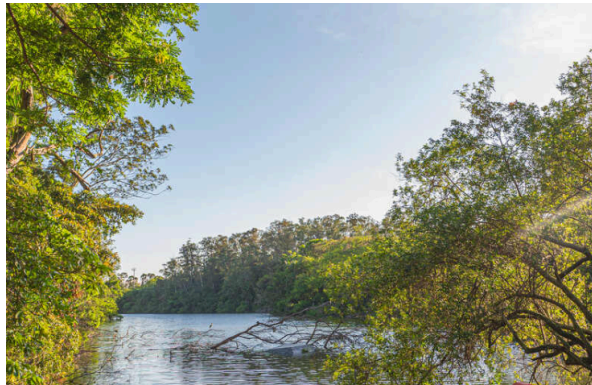


Figure 29: Representing a group that has evolved from burrowing ancestors, the surface-dwelling scallop *Aequipecten irradians* has developed sensory organs along its mantle edges (tentacles and a series of blue eyes).

- **Internal Structure:** Three-chambered heart; U-shaped kidneys.

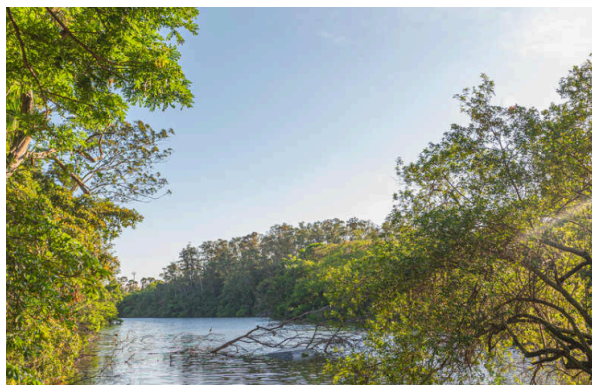


Figure 30: Section through heart region of a freshwater clam to show relation of circulatory and respiratory systems. Respiratory water currents and blood circulation are shown.

Feeding

- Ciliary currents bring water/food to gills.
- Mucus entangles particles; cilia move mucous masses to food grooves and **labial palps**.
- Palps sort particles and direct food to mouth.

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Figure 31: A, Feeding mechanism of freshwater clam. Left valve and mantle are removed.
B, Clam anatomy.

- **Shipworms (*Teredo*)** burrow in wood using valves as rasps; symbiotic bacteria digest cellulose.

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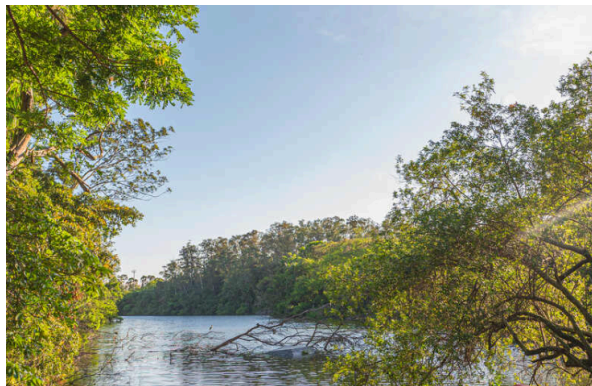


Figure 32: A, Shipworms are bivalves that burrow in wood, causing great damage to unprotected wooden hulls and piers. B, The two small, anterior valves, seen at left, are used as rasping organs to extend the burrow.

- **Giant clams (*Tridacna*)** have symbiotic zooxanthellae in mantle tissue.

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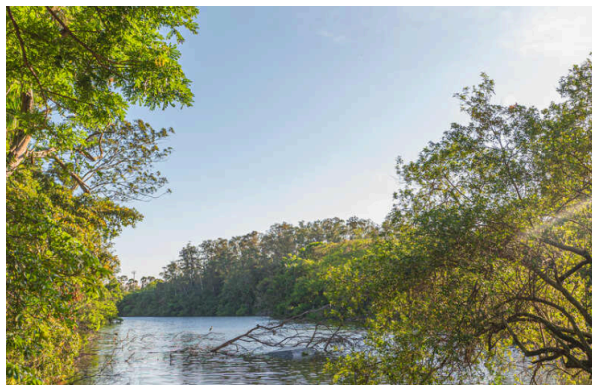


Figure 33: Clam (*Tridacna gigas*) lies buried in coral rock with greatly enlarged siphonal area visible. These tissues are richly colored and bear enormous numbers of symbiotic single-celled algae (zooxanthellae).

- **Crystalline style:** A gelatinous rod in the stomach that rotates to mix food and release enzymes.

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Figure 34: Stomach and crystalline style of ciliary-feeding clam. A, External view of stomach and style sac. B, Transverse section showing direction of food movements. C, Sorting action of cilia.

Reproduction

- Sexes usually separate; external fertilization in marine forms (trochophore, veliger, spat).

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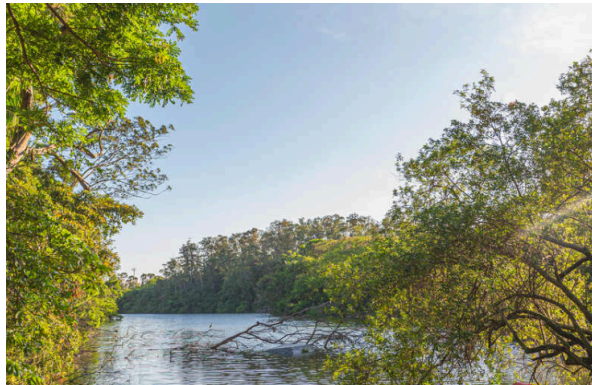


Figure 35: Life cycle of oysters. Oyster larvae swim about for approximately 2 weeks before settling down for attachment to become spats.

- Freshwater clams: Internal fertilization; larvae develop into **glochidia** (parasitic veligers) that attach to fish gills.
- Some mussels have mantle lures to attract host fish.

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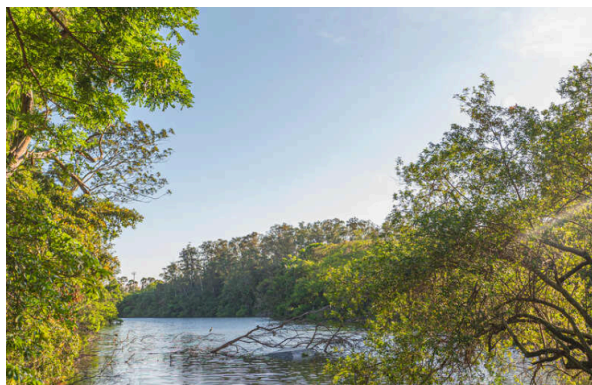


Figure 36: A, Glochidium, or larval form, for some freshwater clams. B, Some clams have adaptations that help their glochidia find a host. The mantle edge of this female pocketbook mussel (*Lampsilis ovata*) mimics a small minnow.

Class Cephalopoda

- Squids, octopuses, nautiluses, cuttlefish.

- All marine active predators.
- Foot modified into a **funnel** (siphon) and a circle of **arms** or **tentacles**.

Shell

- ***Nautilus***: Coiled shell with gas-filled chambers connected by a **siphuncle** (living tissue cord) for buoyancy regulation.
- **Ammonoids** (extinct): Elaborate chambered shells.



Figure 37: *Nautilus*, a cephalopod. A, Live *Nautilus*, feeding on a fish. B, Longitudinal section, showing gas-filled chambers of shell, and diagram of body structure. C, Longitudinal section through shell of an ammonoid.

- **Cuttlefish**: Internal shell (**cuttlebone**).

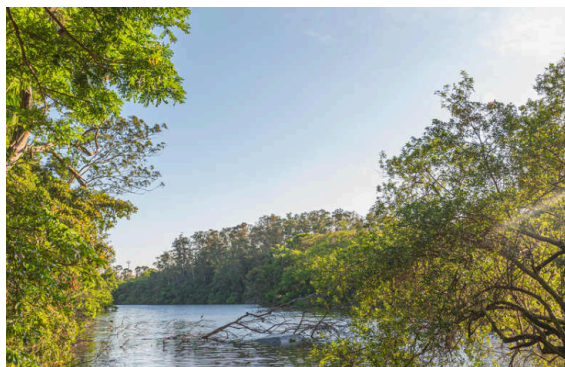


Figure 38: Cuttlefish, *Sepia latimanus*, has an internal shell familiar to keepers of caged birds as “cuttlebone.”

- **Squid**: Thin internal proteinaceous strip (**pen**).
- **Octopus**: Shell absent.

Locomotion and Internal Features

- Swim by jet propulsion (expelling water through funnel).
- **Respiration**: One pair of gills (except *Nautilus* with two); no cilia; muscular pumping of mantle.
- **Circulation**: Closed circulatory system; **branchial hearts** (accessory hearts) at base of gills increase pressure.

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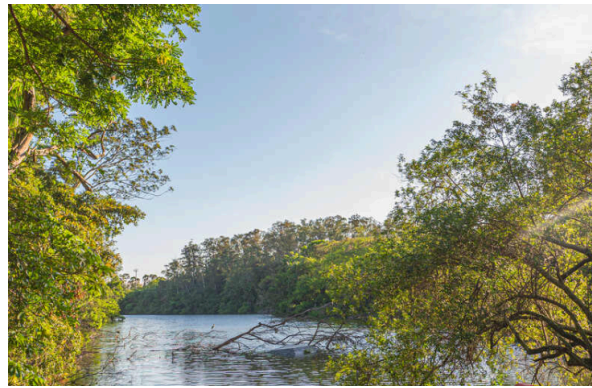


Figure 39: A, Lateral view of squid anatomy, with the left half of the mantle removed. B, Reef squid *Sepioteuthis lessoniana*.

- **Nervous System:** Highly developed brain; giant nerve fibers in squids for rapid escape.
- **Senses:** Complex eyes (cornea, lens, retina) similar to vertebrates (**convergent evolution**).

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Figure 40: Eye of a cuttlefish (*Sepia*). The structure of cephalopod eyes is very similar to that of eyes of vertebrates.

- **Communication:** Visual signals using **chromatophores** (pigment cells) for color changes; ink sac for defense.

Reproduction

- Sexes separate.
- Males use a modified arm (**hectocotylus**) to transfer spermatophores to the female.
- Direct development (no free-swimming larva).

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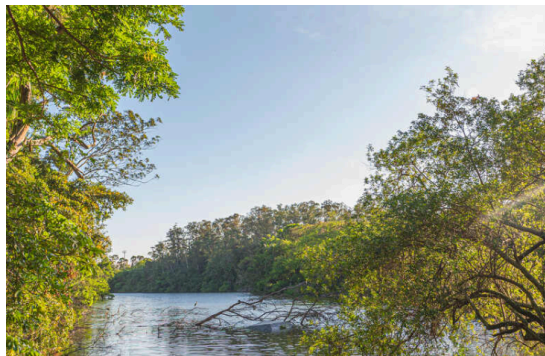


Figure 41: Copulation in cephalopods. A, Mating cuttlefishes. B, Male octopus uses modified arm to deposit spermatophores in female mantle cavity to fertilize her eggs.

Major Groups

- **Nautiloidea**: *Nautilus* (2 pairs of gills).
- **Ammonoidea**: Extinct.
- **Coleoidea**: 1 pair of gills.
 - **Sepioidea** (cuttlefish): 8 arms, 2 tentacles.
 - **Teuthida** (squids): 8 arms, 2 tentacles.
 - **Octopoda** (octopuses): 8 arms, no tentacles.

Phylogeny and Adaptive Diversification

- Molluscs are lophotrochozoan protostomes.
- Relationship with annelids is debated (shared trochophore, but lack of segmentation in ancestral mollusc).
- “Hypothetical Ancestral Mollusc” likely wormlike with dorsal mantle and scales.
- Adaptive radiation driven by mantle and shell modifications.
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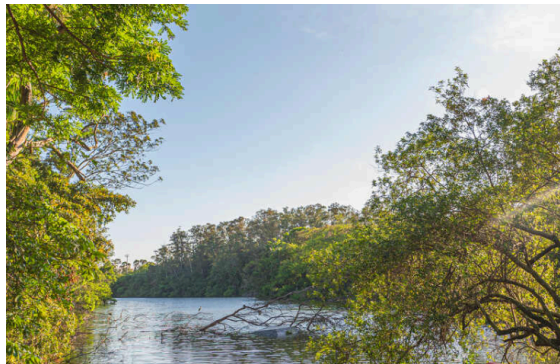


Figure 42: Classes of Mollusca, showing their derivations and relative abundance.